DataLex: Explainable SIEM Security Intelligence Using Generative AI

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SCOPES

01

INTRODUCTION

INTRODUCTION

- In the digital transformation age, cybersecurity has become a critical frontier.

- Cyber threats have increased due to the exponential rise in digital infrastructures.

What is SIEM?

Security Information and Event Management (SIEM)

- A system that collects, aggregates, and analyzes security logs.
- Used for real-time monitoring and threat detection.
- Helps security teams respond to incidents quickly.

Basically it’s an umbrella under which all logs are aggregated.

Ex. Splunk, Fortinet, IBM qRadar

PROBLEM STATEMENT

- Cost is too high for many companies in Bangladesh
- Lacks contextual understanding of interconnected security events
- Overwhelming dependence on external cloud-based API services
- Significant privacy risks from third-party data transmission
- Inability to provide human-readable, actionable threat intelligence

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PROBLEM STATEMENT & MOTIVATION

MOTIVATION

Why We Introduced DataLex

Bridging the Gap:

- Traditional SIEM lacks explainability and AI-driven automation.
- Security analysts struggle with inefficient threat detection methods.
- Cloud-based SIEMs create privacy concerns.
- Initially the solution can work as Threat Intel as well to some extent.

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PROPOSED SOLUTION

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- Lower false positives and clear contextual understanding.
- No dependency on external cloud-based services for AI processing.
- Scalable, explainable, and adaptive security solutions.
- Plug and Play (PnP) technology

Introducing DataLex

- An AI-powered, offline, and explainable SIEM.
- Uses LLAMA 3.0 and DeepSeek Qwen for intelligent analysis. Users can choose which model they want to use.
- Privacy-preserving threat detection with zero external API reliance.
- Through Chat interface, users can interact.

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SYSTEM DESIGN

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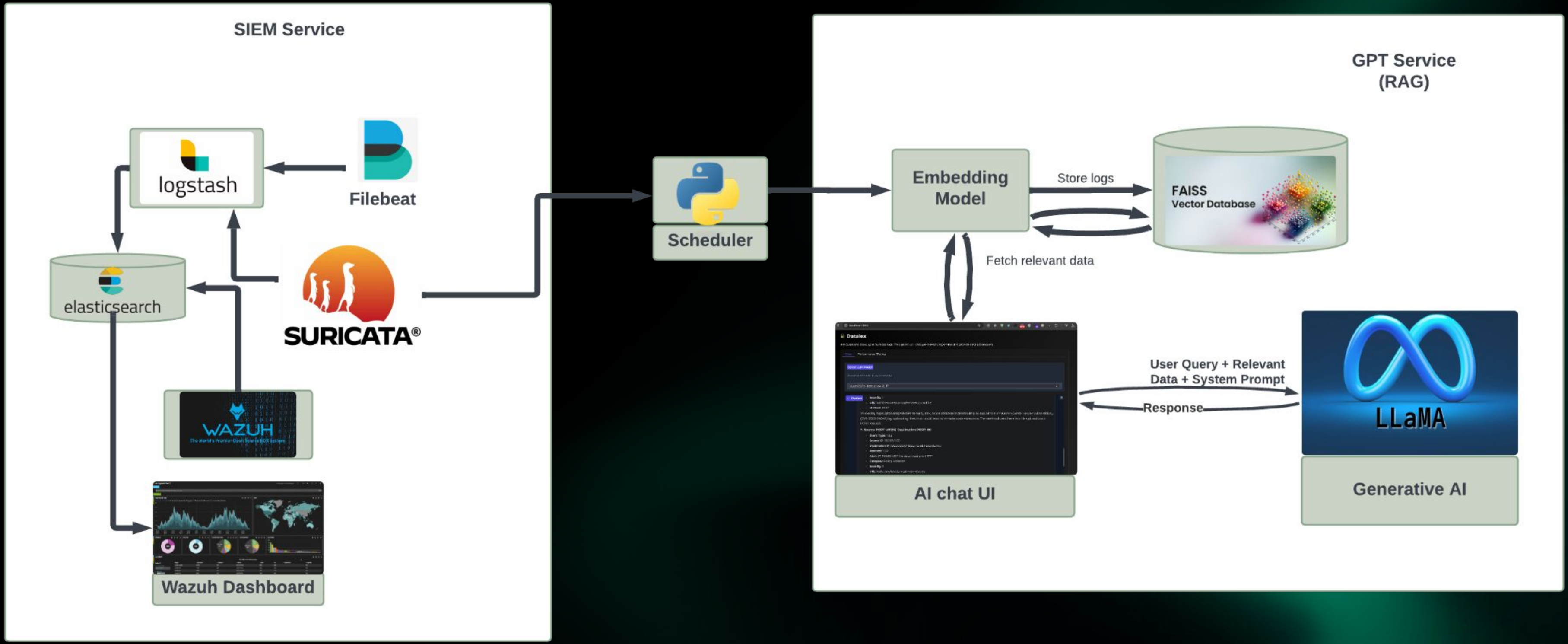

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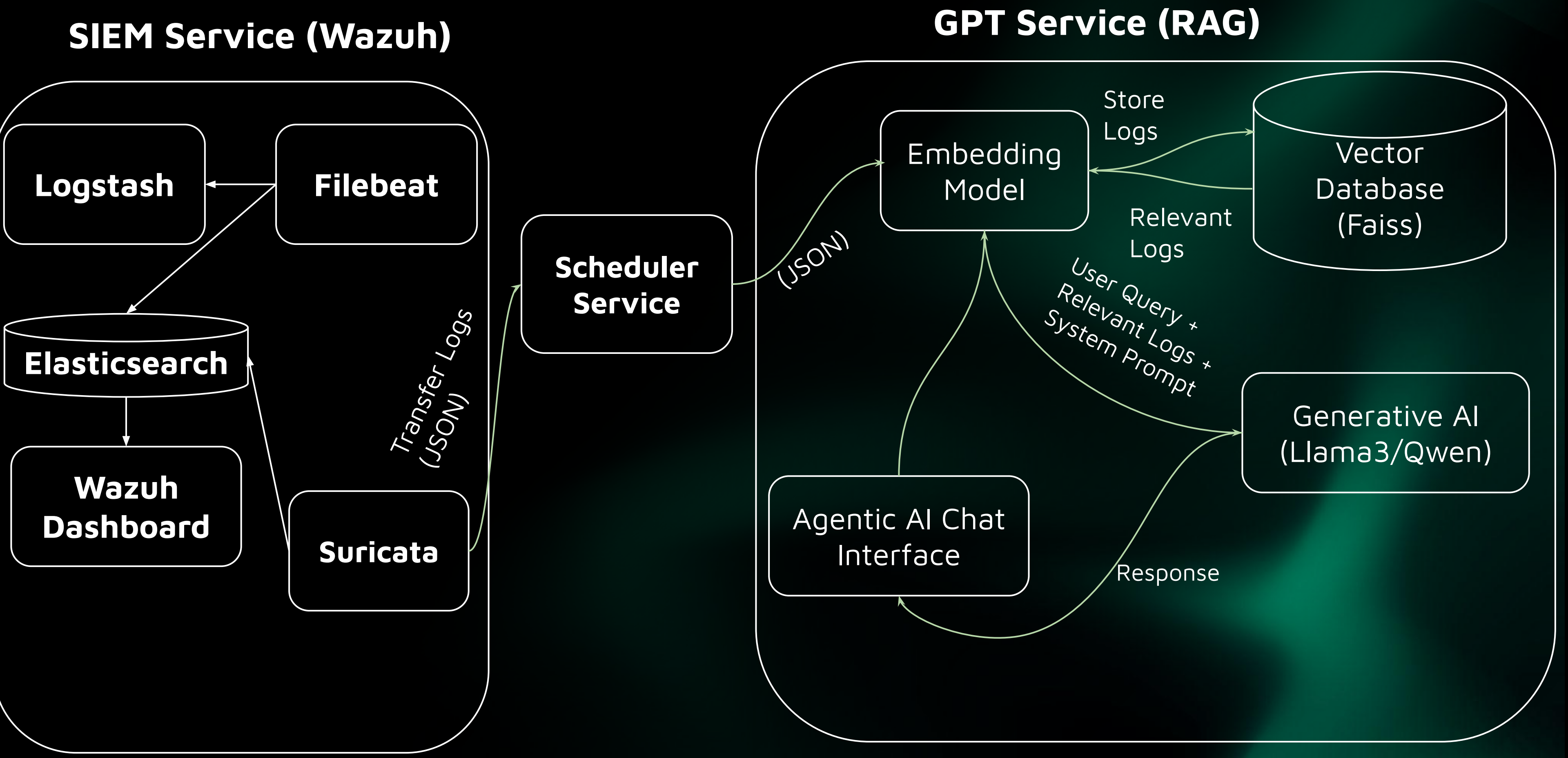
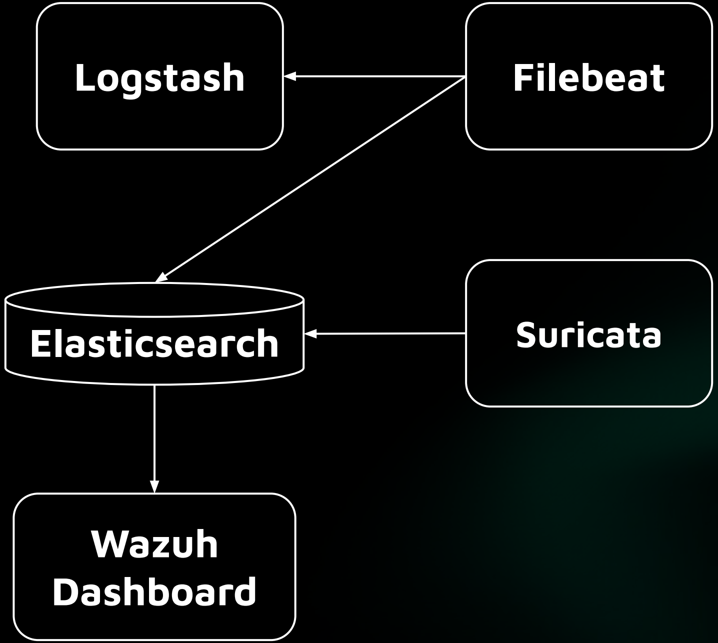
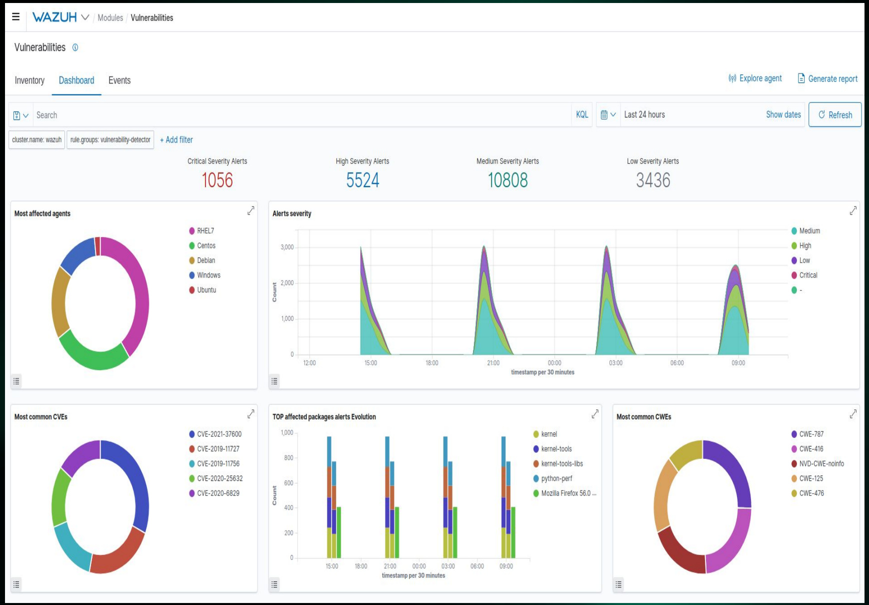


SYSTEM DESIGN

SIEM Service (Wazuh)





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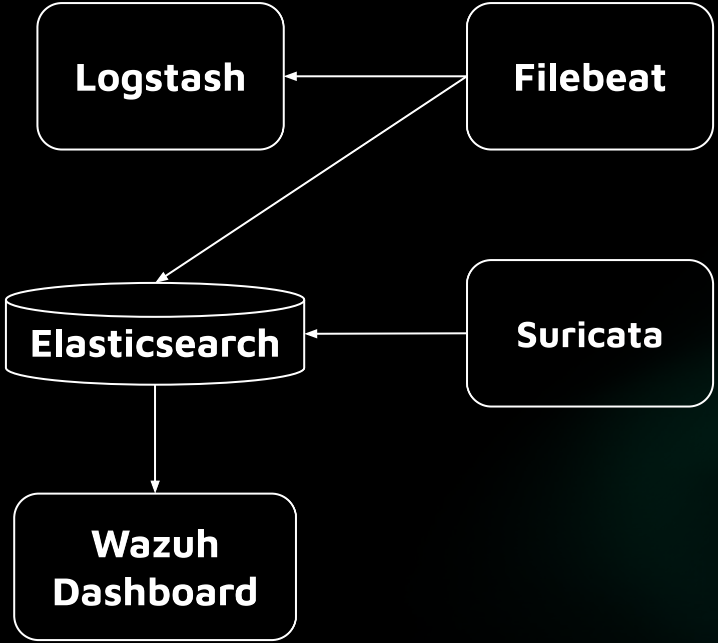
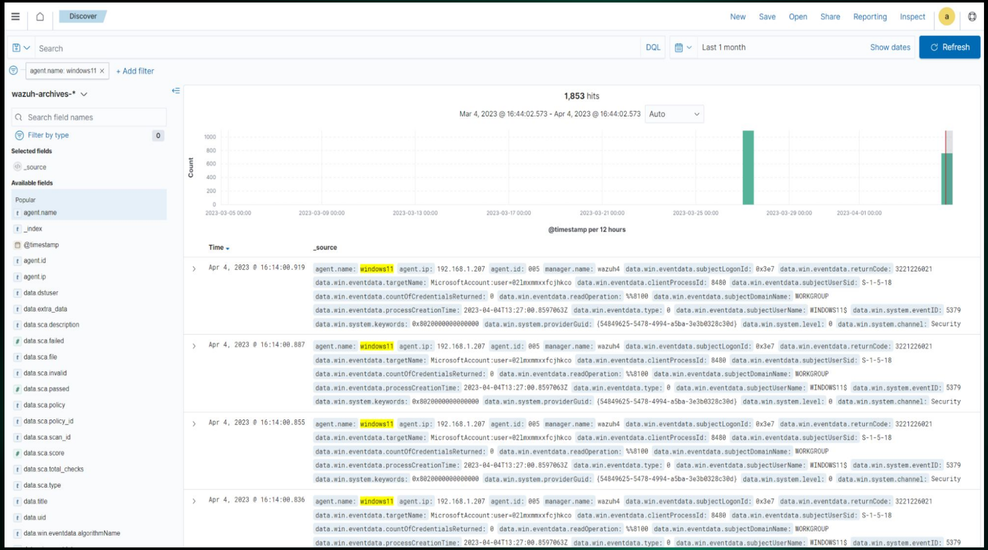




SYSTEM DESIGN

SIEM Service (Wazuh)

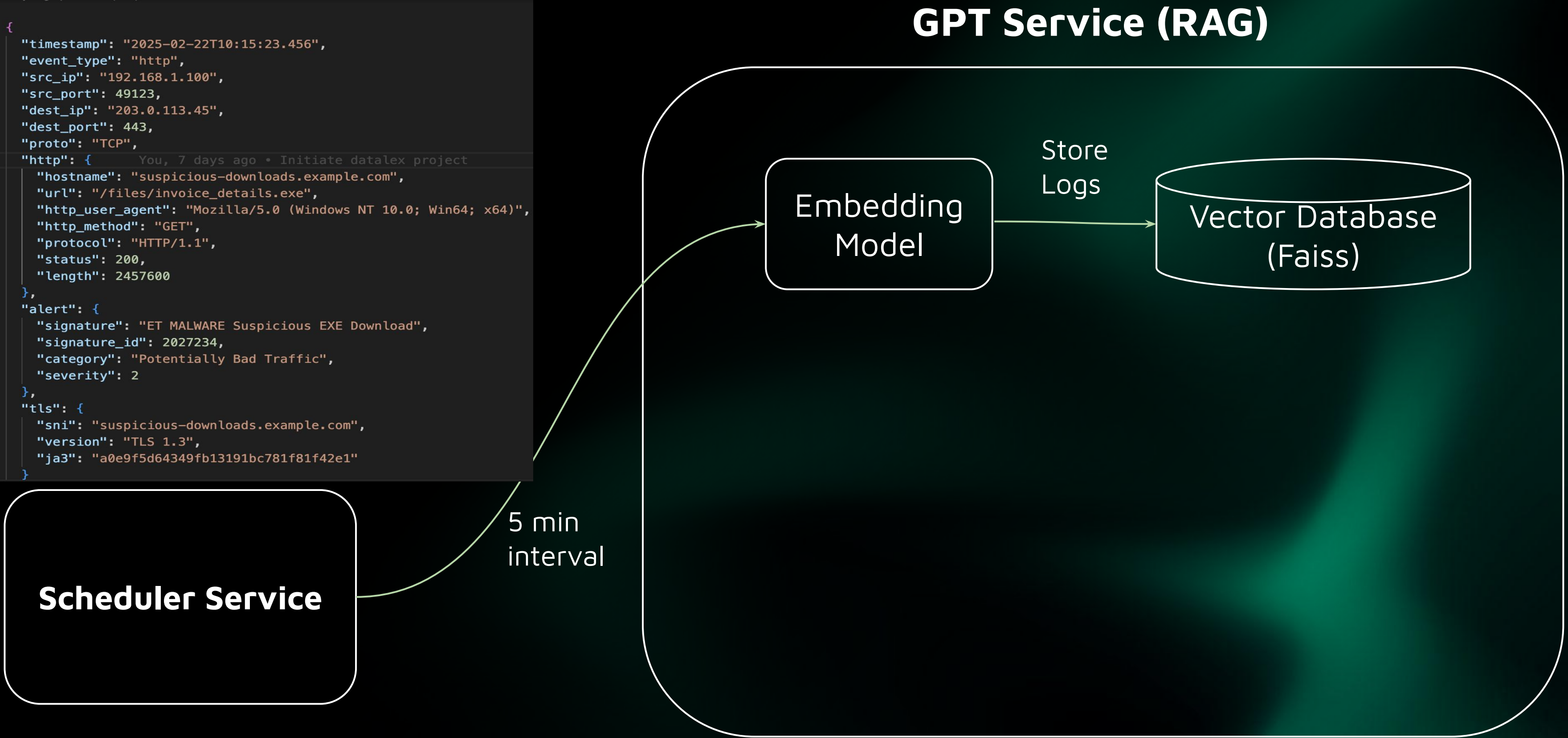

```
mac@sajid datalex % cd scheduler-service
mac@sajid scheduler-service % source venv/bin/activate
(venv) mac@sajid scheduler-service %  python app.py
INFO:     Will watch for changes in these directories: ['/Users/mac/Desktop/workspace/datalex/scheduler-service']
INFO:     Uvicorn running on http://localhost:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [16005] using WatchFiles
INFO:apscheduler.scheduler:Scheduler started
INFO:root:-------------CRON JOB INITIATED-------
INFO:apscheduler.scheduler:Added job "job_wrapper" to job store "default"
INFO:     Started server process [16007]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:apscheduler.executors.default:Running job "job_wrapper (trigger: interval[0:01:00], next run at: 2025-03-05 01:11:41 +06)" (scheduled at 202
INFO:root:Starting transfer_suricata_logs job.
=================SURICATA LOG TRANSFER INITIATING===========
Successfully cloned and transformed JSON into a list: /Users/mac/Desktop/workspace/datalex/gpt-service/data/eve.json
=================SURICATA LOG TRANSFER FINISHED===========
INFO:root:Finished transfer_suricata_logs job.
INFO:apscheduler.executors.default:Job "job_wrapper (trigger: interval[0:01:00], next run at: 2025-03-05 01:12:41 +06)" executed successfully
_
```

SYSTEM DESIGN

Suricata Logs

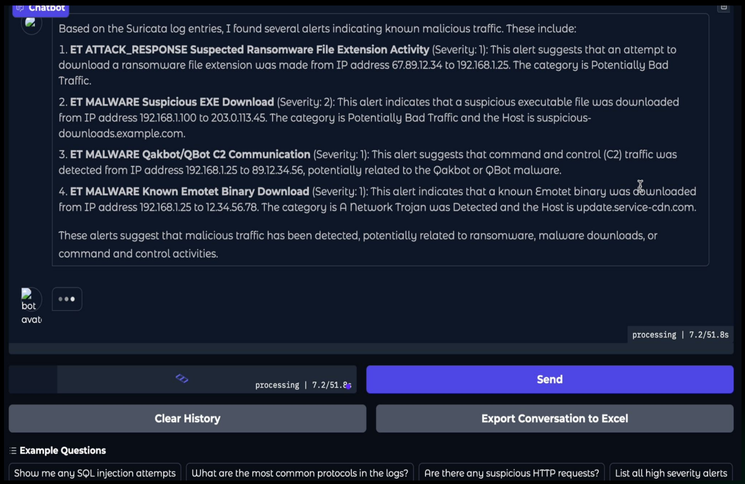
gpt-service > data > {} 0_to_4_suricata.json > {} 0 > {} http

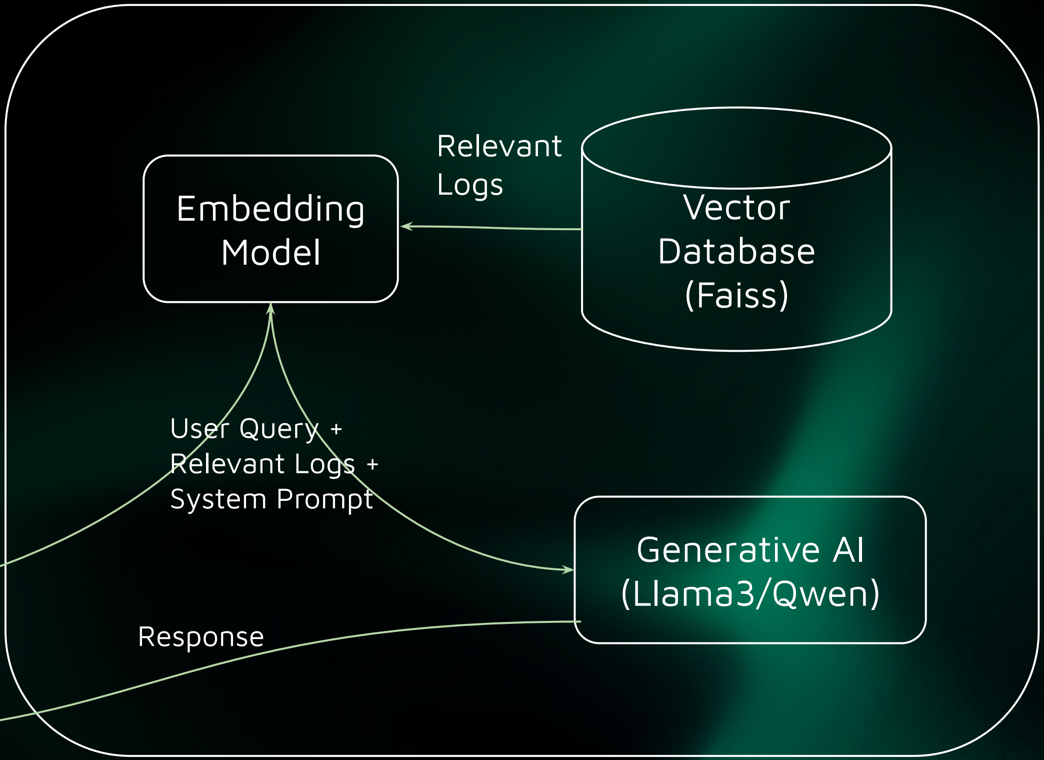

```json
[
    {
        "timestamp": "2025-02-22T10:15:23.456",
        "event_type": "http",
        "src_ip": "192.168.1.100",
        "src_port": 49123,
        "dest_ip": "203.0.113.45",
        "dest_port": 443,
        "proto": "TCP",
        "http": {
            "hostname": "suspicious-downloads.example.com",
            "url": "/files/invoice_details.exe",
            "http_user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64)",
            "http_method": "GET",
            "protocol": "HTTP/1.1",
            "status": 200,
            "length": 2457600
        },
        "alert": {
            "signature": "ET MALWARE Suspicious EXE Download",
            "signature_id": 2027234,
            "category": "Potentially Bad Traffic",
            "severity": 2
        },
        "tls": {
            "sni": "suspicious-downloads.example.com",
            "version": "TLS 1.3",
            "ja3": "a0e9f5d64349fb13191bc781f81f42e1"
        }
```

SYSTEM DESIGN

GPT Service (RAG)





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TECH STACK

TECH STACK

- Language/Framework: Python, FastAPI.
- Log Management: Wazuh, Logstash, Filebeat
  Elasticsearch
- Visualization: Wazuh Dashboard.
- Intrusion Detection: Suricata.

TECH STACK

- AI/ML: Ollama, LLAMA3/Qwen, FAISS, Gradio
- Containerization: Docker for scalable and isolated service deployment
- Threat Intelligence: Virus Total

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RESULT ANALYSIS

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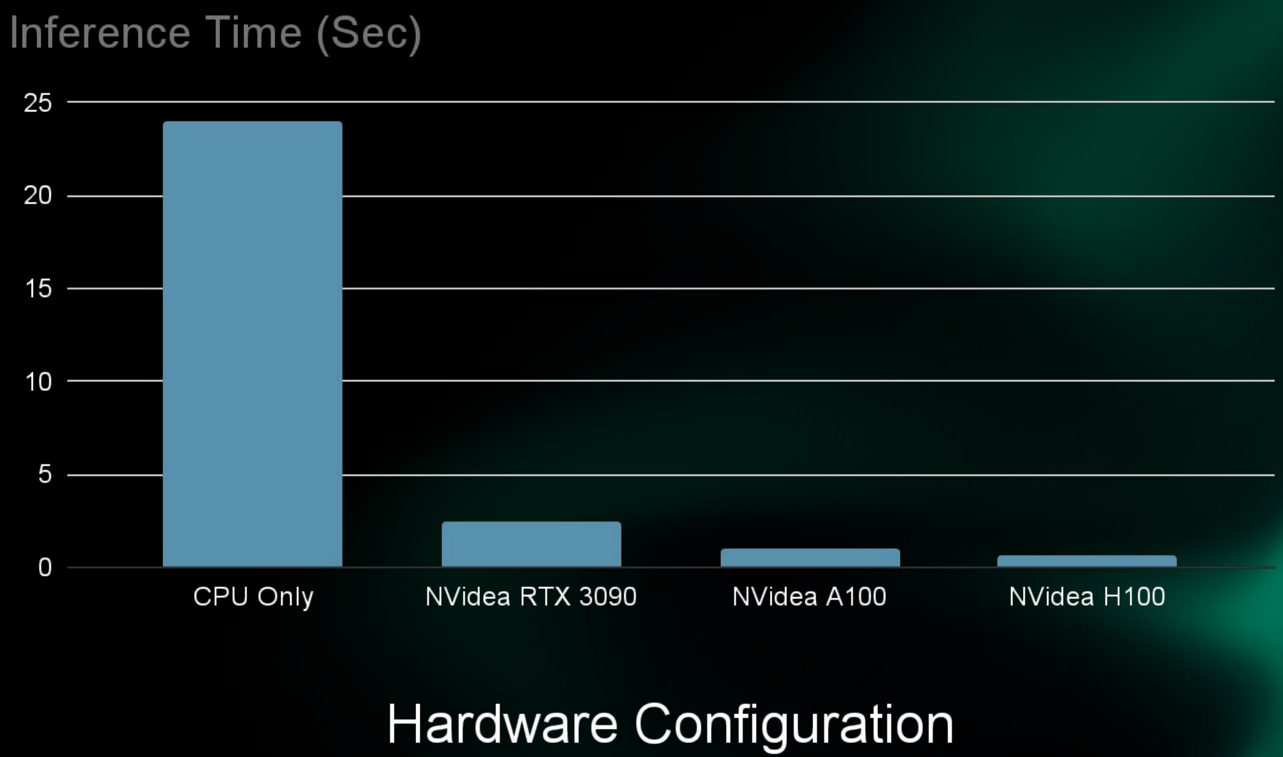
- Inference Time = Retrieval Latency + Generative AI response
- Retrieval Latency = Required time to fetch logs from vector database
- Memory Usage = Required memory for processing per query
- User Query/Min = Number of query execution per minute
- Human Labeling on conversation logs

RESULT ANALYSIS



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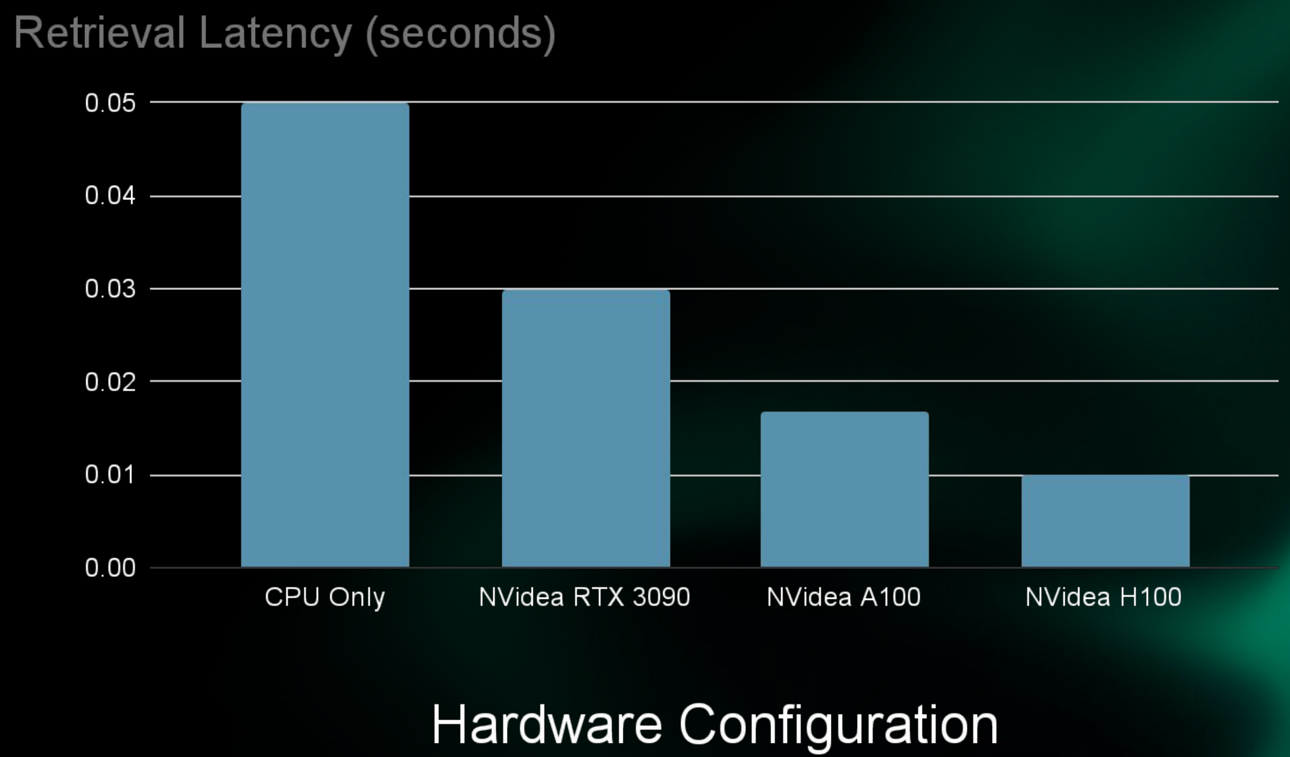

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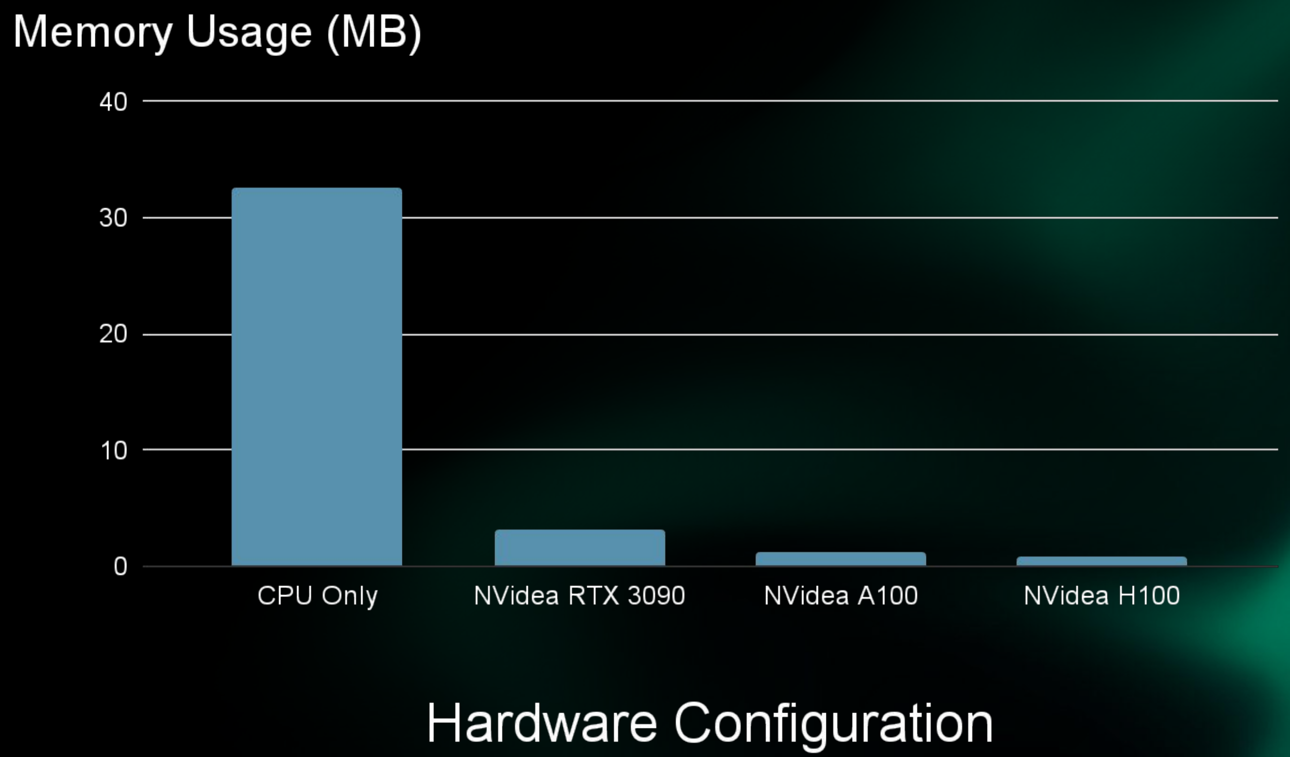

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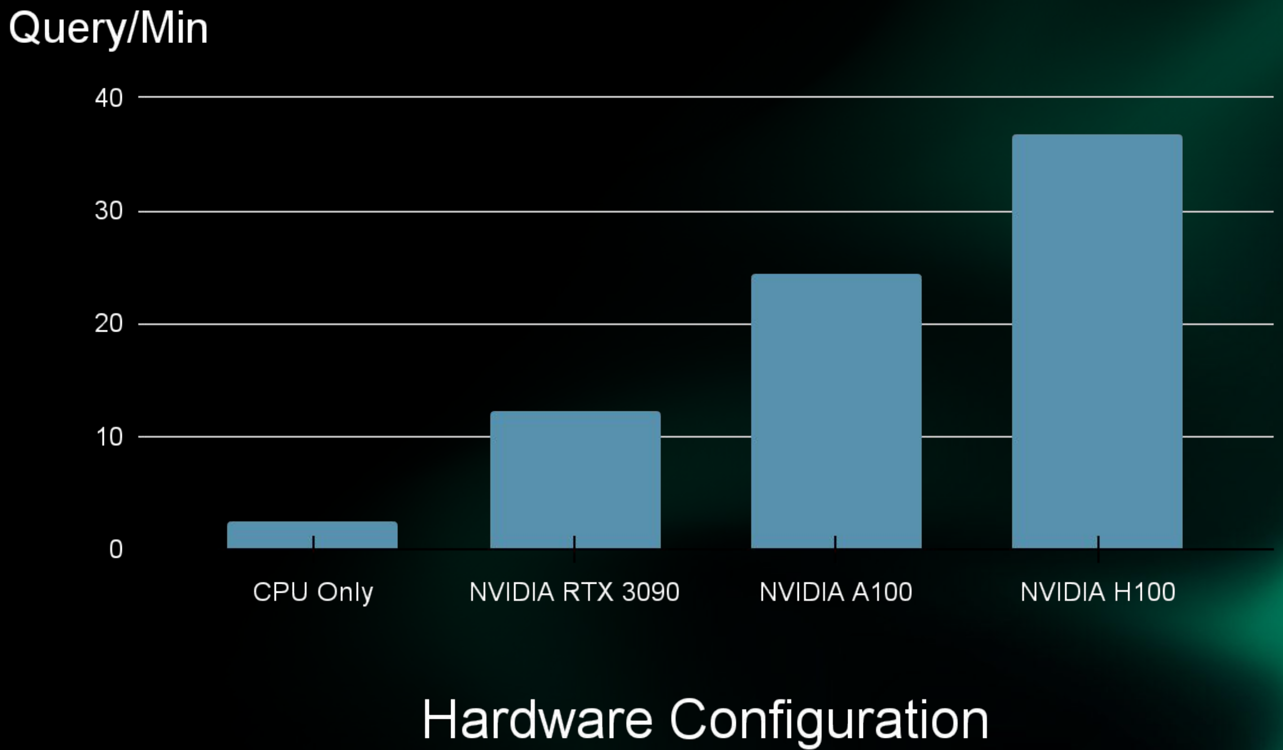
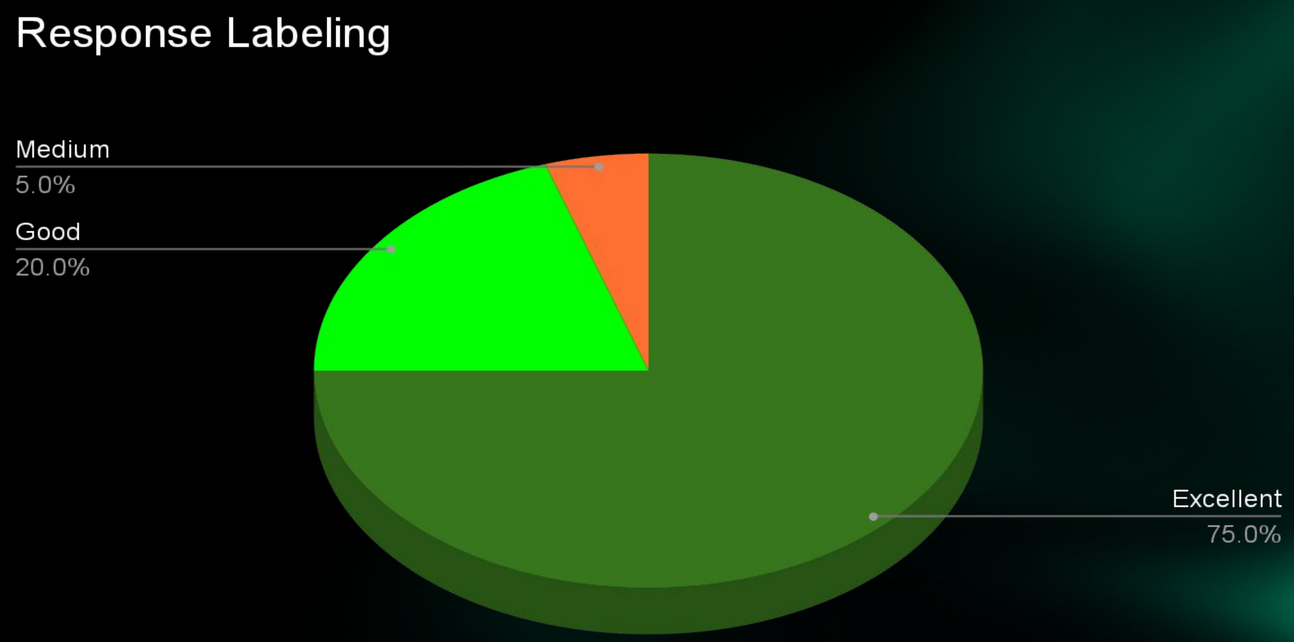


RESULT ANALYSIS

Conversation Logs

Question	Response
Show me any SQL injection attempts	A thorough analysis of the Suricata log entries! After examining the logs, I did not find any SQL injection attempts. The alerts and events in the logs suggest various types of attacks, including 1. An attempt to exploit the Apache Log4j RCE (Remote Code Execution) vulnerability (CVE-2021-44228). 2. A suspicious EXE download from a potentially malicious source. 3. Attempted exploitation of MSBlaster Worm Transfer and Windows RDP BlueKeep RCE vulnerabilities. 4. An anomaly detected in an HTTP GET request for a critical update executable file. However, I did not find any log entries indicating SQL injection attempts. The logs do not show any patterns or characteristics of SQL injection
What are the most frequent types of alerts generated by Suricata in the last 24	Based on the Suricata log entries, the most frequent types of alerts generated in the last 24 hours are: 1. **Potentially Bad Traffic** (Severity: 1) - This alert type is triggered by multiple events: * Source PORT: 445 | Destination PORT: 49872 | Event Type: alert | ... | Alert: ET ATTACK_RESPONSE Suspected Ransomware File Extensio * Source PORT: 49874 | Destination PORT: 6892 | Event Type: alert | ... | Alert: ET MALWARE Known Emotet Binary Download * Source PORT: 49123 | Destination PORT: 443 | Event Type: http | ... | Alert: ET MALWARE Suspicious EXE Download These alerts indicate potential malicious activity, such as ransomware and malware downloads. 2. **Remote Code Execution** (Severity: 1) - Triggered by one event: * Source PORT: 44552 | Destination PORT: 80 | Event Type: alert | ... | Alert: ET EXPLOIT Apache Log4j RCE Attempt (CVE-2021-44228) This alert indicates an attempted exploitation of the Apache Log4j RCE vulnerability. 3. **Command and Control Traffic** (Severity: 1) - Triggered by one event: * Source PORT: 49873 | Destination PORT: 443 | Event Type: alert | ... | Alert: ET MALWARE Qakbot/QBot C2 Communication

RESULT ANALYSIS



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COMPETITOR ANALYSIS

COMPETITOR ANALYSIS

Feature	Datalex	Splunk	Fortinet
Offline AI Processing	✅	❌	❌
More Control over AI	✅	❌	❌
Support for Multiple AI models	✅	❌	❌
Support for Bangla Language	✅	❌	❌

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FUTURE WORKS

FUTURE WORKS

- Need more fine tune of RAG for better outputs.
- Add support for custom knowledgebase.

Thanks!

Do you have any questions?

REFERENCE

- AI Test Case: conversation_log_20250301_024007.xlsx
- [GPU Requirement Guide for LLaMA3](https://apxml.com/posts/ultimate-system-requirements-llama-3-models)
- [Building a LLaMA 3.2-11B-Based RAG System](https://www.e2enetworks.com/blog/building-a-llama-3-2-11b-based-rag-system-with-vision-model-integration)
- [Wazuh Server Installation Guide](https://documentation.wazuh.com/current/installation-guide/wazuh-server/index.html)
- [Elastic Stack Integration with Wazuh](https://documentation.wazuh.com/current/integrations-guide/elastic-stack/index.html)